

CSE400: Fundamentals of Probability in Computing

Milestone 4 Scribe

Handover Probability in Drone Cellular Networks

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Scribe Question 1: Baseline System and Motivation for Randomization

What was the deterministic baseline?

In Milestone 3, we built a simulation of a drone cellular network where UAVs act as flying base stations serving a single stationary user on the ground. The drones' starting positions were placed using a Poisson Point Process at a fixed density, and each drone then travels in a straight line at a random angle.

The system always connects the user to whatever drone happens to be physically closest at that moment — this is the nearest-neighbor association policy:

$$x^*(t) = \arg \min_i \|x_i(t)\|$$

A handover gets recorded every time the identity of that closest drone changes. We looked at two movement scenarios: SSM, where every drone flies at the same fixed speed, and DSM, where each drone is assigned its own speed sampled from either a Rayleigh or Uniform distribution. Every parameter in M3 was held constant, matching the values confirmed in the paper and slides:

Parameter	M3 Fixed Value
Drone density λ_0	1.0 DBS/km ² = 1×10^{-6} /m ²
Speed (SSM)	45 km/h (12.5 m/s)
Exclusion radius u_0	500 m
Time window T	400 s
Monte Carlo runs	1,00,000 simulations

Why introduce randomization?

Running everything at one fixed set of values only tells us the theory holds for that single scenario. In practice, drone networks look different every time they're deployed — the number of drones, their speeds, and the distances involved all vary. When we instead draw parameters randomly each run, every simulation becomes its own independent test. If the theoretical predictions still hold up across all of those varied runs, we can be confident the results genuinely generalize rather than being a coincidence of one carefully chosen configuration.

Scribe Question 2: Randomized Algorithm Design

What component now involves randomness?

We built a new central module called **randomised_config.py** that executes at the very beginning of each simulation. Rather than loading hard-coded constants, it samples five key physical parameters fresh from Uniform distributions every run. Everything else — the nearest-neighbor connection rule, the handover detection logic, and how the drones move — stays exactly the same as it was in M3.

What distributions are used?

Parameter	Distribution	Range (from code)
Drone density λ_0	Uniform	5×10^{-7} to 2×10^{-6} UAVs/m ²
Mean speed v^{Mean}	Uniform	30 to 60 km/h
Speed spread v^{Diff}	Uniform	20 to 120 km/h
Exclusion radius u_0	Uniform	300 to 800 m
Time window T	Uniform (integer)	200 to 400 s

The specific values that were actually drawn for the run used to produce our graphs — verified against the saved JSON file, were:

Parameter	M3 Deterministic (Fixed)	M4 Randomized (Actual Draw)
λ_0	1.00×10^{-6} /m ²	1.44×10^{-6} /m ²
Mean speed	45.0 km/h	53.1 km/h
u_0	500 m	335.8 m
T	400 s	272 s
Simulations	1,00,000	10,000

How does it differ from the deterministic baseline?

In M3, the exact same numbers fed into every single run. In M4, each run is genuinely a fresh experiment with its own parameter set. To keep the comparison fair, a **save-and-load system** is used: the sampled values are written to a JSON file immediately, and every theory script then reads from that same file, so the simulation and the theoretical calculations are always working from identical inputs.

Scribe Question 3: Probabilistic Reasoning Behind the Algorithm**How does randomness help address uncertainty?**

In a real-world drone network, nothing stays fixed — drone counts, speeds, and deployment distances all shift from one situation to the next. If we only ever test at one specific configuration, the most we can conclude is that the theory works there. By sampling parameters from distributions, each run probes the theory at a completely different operating point. Agreement across all of those varied conditions is a much stronger form of validation.

What probabilistic models support this design?

- **PPP for drone deployment:** The paper's derivations hold for any density λ_0 , so randomizing it simply tests the same theory at a different point rather than violating any assumption.
- **Displacement Theorem (Lemma 1):** This result guarantees that drones remain Poisson-distributed at every instant regardless of how they move, for any λ_0 — our randomized value still falls within that guarantee.
- **Speed PDF for DSM (Lemma 3):** Lemma 3 applies to any valid speed distribution $f_v(v)$, so choosing a different Rayleigh sigma σ just plugs a different valid distribution into the same formula without breaking anything.
- **Exclusion radius u_0 :** Varying u_0 across the 300–800 m range captures the reality that drones don't always land at the same distance from a user.

Why is this expected to improve the analysis?

The randomized approach doesn't change the underlying handover physics. What it does is make our conclusions far more defensible. A finding that holds across many independently drawn parameter settings is substantially more convincing than one that was only ever verified at a single hand-picked configuration.

Scribe Question 4: Implementation Within the Simulation Pipeline

What changes were made to the code?

The only new file created in this instance of the code is `randomised_config.py`. It has three functions: `sample_params()`, which randomizes and gets all five parameter values; `save_params()`, which saves the five parameters to a JSON file; and `load_params()`, which loads the five parameters back from the JSON file. All simulation and theory scripts call `load_params()` so that they will all use the same sampling of parameters.

The updated pipeline is as follows:

- **Step 1:** Call `sample_params()` in the `handover_dbs.simulation.py` script. Samples the five parameters, drawn from their distributions.
- **Step 2:** Displays the five parameters just sampled. All those parameters are saved in `Data/handover_run_params.json`.
- **Step 3:** 10,000 Monte Carlo simulations are run using the sampled values for λ_0 , $V_{\min}$, $V_{\max}$, and T ; where the parameter λ_0 is sampled using random values.
- **Step 4:** All theory scripts that simulate constant velocity, the Rayleigh lower bound and the Uniform lower bound will call `load_params()` and retrieve their parameters using the same JSON file.
- **Step 5:** The script `plot_handover.py` loads all the result `.npz` files and generates the final comparative plot.

How is randomness generated inside each realization?

- Number of drones: $N \sim \text{Poisson}(\lambda_0 \times \pi \times R^2)$, $R = 30$ km.
- Starting positions: $r = R \times \sqrt{U}$, $U \sim \text{Uniform}(0, 1)$; angle $\theta \sim \text{Uniform}(0, 2\pi)$.
- Movement direction: $\varphi \sim \text{Uniform}(0, 2\pi)$ per drone.
- DSM speeds: $v \sim \text{Uniform}(0, 107.5 \text{ km/h})$ per drone for this run.
- SSM speed: all drones share the same constant $v_{\text{Mean}} = 53.1 \text{ km/h}$ for this run.

Does everything else in the pipeline remain unchanged?

Yes, all other aspects regarding the nearest-neighbor decision process, how to detect when a user is handed over, how to keep a drone moving straight, and the Monte Carlo loop remain unchanged from M3 with regard to their mechanics; however, instead of being hard-coded, the 5 input parameters will now be randomly drawn from the input generation distribution.

Scribe Question 5: Evaluation Metrics and Comparative Results**Why are these metrics appropriate for this problem domain?**

We used 4 metrics that connect to each other in order to gauge if the randomized DSM is superior to the deterministic SSM system. Both M3 and M4, which evaluate the two systems, utilize the same nearest-neighbour rule as well as the same simulation process; only the set of parameters would differ between M3 and M4.

The four metrics we created are:

- (a) Handover Probability $P[H(t)]$ — the foundation metric of the domain
- (b) Expected Communication Cost — resource utilization metric
- (c) Accumulated Latency (Service Interruption) — user experience metric
- (d) Connection Stability $CS(t) = 1 - P[H(t)]$ — stability metric

Confirmed parameter values used in each approach:

Parameter	M3 Deterministic (paper + PPT)	M4 Randomized (saved JS)
λ_0	1.00/km ² ($1 \times 10^{-6}/\text{m}^2$)	1.44/km ² ($1.44 \times 10^{-6}/\text{m}^2$)
Speed	45 km/h (fixed every run)	53.1 km/h (drawn randomly t
u_0	500 m	335.8 m
T	400 s	272 s
Simulations	1,00,000	10,000

Metric 1: Handover Probability $P[H(t)]$ — Foundational Domain-Specific Measure

What it is: The likelihood that a ground user will have had to switch to a different drone connection at least once by time t . This is the primary metric for the project as a whole and is based upon Theorem 2 (SSM) and Theorem 3 (DSM) as defined in the Theoretical Paper.

Why it is appropriate: Handover Probability is a direct measure of the stability of the connection between a user and the drone cellular network. All other metrics that have been defined will be based upon this metric. A lower $P[H(t)]$ will necessarily be better than a higher $P[H(t)]$, since it indicates that a user has maintained their original connection to their drone longer than alternative connections.

From the M3 Handover Simulation (Deterministic, 100,000 simulations): From the results obtained from running the simulation with 100,000 runs, the 3 deterministic curves (Rayleigh, Constant, and Uniform 40) have a common characteristic in that they increase steeply from 0. The simulation points from the 100,000 runs fall on top of the theoretical traces for all of the curves from 0 through 300 m/s. Constant speed and Uniform 40 had both reached a value of $P[H] \approx 0.97$ by $u_x = 150$ m.

Rayleigh lagged well behind, achieving its approximate maximum (i.e. ≈ 1.0) by $u_x = 300$ m. These two measures demonstrate that there is an extremely close correspondence between the Theoretical and Simulation results at the same fixed deterministic variables.

Regarding the transfer of probability from the M4 graph (Randomized, SSM vs DSM) in terms of time t : From the M4 graph above, the x -axis is a function of time (t). SSM and DSM both start at approximately 0.05 at $t = 0$ and remain similarly close until approximately 40 s. At this point, however, SSM increases rapidly compared to DSM which continues to develop slowly. In addition, the Mean Squared Error (MSE) between the simulated and theoretical lower bound to that of DSM is only 6.2×10^{-6} which indicates that the theory remains valid under both initialized iterations.

Threshold	M3 Det. SSM (approx.)	M4 Ran. SSM	M4 Ran. DSM
$t_{50\%}$ (50% HO)	38 s	25 s	24 s
$t_{90\%}$ (90% HO)	69 s	60 s	104 s
$t_{99\%}$ (99% HO)	97 s	91 s	Never in 272 s

Significant Finding: DSM reaches a 90% (0.9) handover probability by $t = 104$ s and SSM achieves the same level of handover by $t = 60$ s, a 44-second difference in active real-time connection duration. Additionally, SSM achieves 99% (0.99) by 91 s while DSM does not achieve 99% (0.99) during the complete simulation period; consequently, Randomized DSM users remain associated with their original UAV devices longer than Deterministic SSM users.

Metric 2: Expected Communication Cost — Resource Utilization Metric

What This Is: In a cellular drone network, whenever a user hands over from one drone to another, control signals (i.e., messages for authenticating the user, requesting resource allocations, and transferring sessions, also known as signaling overhead) have to be exchanged between the user, old drone, and new drone before the handover can take place. Therefore, Handover Probability can serve as a direct representation of this cost (i.e., the lower $P[H(t)]$, the lower the overhead of signaling and the greater the total amount of bandwidth available for actual user data).

Why This Is Appropriate: As indicated on the M4 rubric, the utilization of all resources (e.g., network) is a valid metric. In a cellular drone network, control signals (e.g., signaling overhead) use bandwidth that could otherwise be used for user data. Therefore, a strategy that reduces handovers will have a more resource-efficient environment.

How We Measure This: The communication cost savings will be defined as a ratio of the number of users that do not trigger signaling for a handover in a DSM compared to an SSM:

$$\text{Cost Saving}(t) = P_{\text{SSM}}(t) - P_{\text{DSM}}(t)$$

Time t	SSM Cost P_{SSM}	DSM Cost P_{DSM}	Cost Saving (SSM–DSM)
$t = 30$ s	0.582	0.597	−0.015 (early on, nearly equal)
$t = 50$ s	0.828	0.771	+0.057 (5.7% fewer handovers in)
$t = 60$ s	0.903	0.820	+0.083 (8.3% saving)
$t = 88$ s	0.987	0.879	+0.108 (10.8% saving — peak)
$t = 100$ s	1.000	0.895	+0.101 (10.1% saving)
$t = 150$ s	1.000	0.930	+0.070 (7.0% saving)
$t = 200$ s	1.000	0.949	+0.051 (5.1% saving)

Observation: The maximum possible savings found in communicating was 10.8% at $t \approx 88$ s into the test. This means that, at the time where communication efficiency was maximized, SSM users were generating 10.8% more unnecessary handover control signals than DSM users. The randomized approach to DSM has a demonstratively greater efficiency than M3. The M3 deterministic setting showed that all three graphs converged very quickly and there is very little visible separation. However, M4 randomized demonstrates that there is a definitive and significant communication cost efficiency (i.e. DSM) under a variety of different parameter conditions.

Metric 3: Accumulated Latency (Service Interruption) — User Experience Metric

What it is: *A handover is typically not instantaneous in nature. However, a user will always experience a slight data interruption during any type of real 5G network handover due to the requirement for re-authentication with the new base station. This process typically takes 30–50 ms to complete and has been given a midpoint of 40 ms for all handover events in our study. Thus the expected service interruptions at various time intervals.*

$$\text{Expected Latency}(t) = P[H(t)] \times 40 \text{ ms}$$

Why it is appropriate: *The reason this is valid is because latency is defined in the M4 rubric as a valid metric when dealing with real-time applications such as video calls, remote control of drones, and/or streaming from sensors. Therefore even 40 ms of service interruption in between two connections is significant enough to warrant a reduction in total service interruption by reducing the amount of time between handovers through effective handover strategy.*

Time t	SSM Latency	DSM Latency	Latency Saved by DSM
$t = 30$ s	23.3 ms	23.9 ms	−0.6 ms (nearly equal early)
$t = 50$ s	33.1 ms	30.8 ms	+2.3 ms saved
$t = 60$ s	36.1 ms	32.8 ms	+3.3 ms saved
$t = 88$ s	39.5 ms	35.2 ms	+4.3 ms saved (peak)
$t = 100$ s	39.8 ms	35.8 ms	+4.0 ms saved
$t = 150$ s	40.0 ms	37.2 ms	+2.8 ms saved
$t = 200$ s	40.0 ms	38.0 ms	+2.0 ms saved

Key finding: *The key finding was that at the end of $t = 88$ s, a DSM will have seen an overall reduction in total service interruptions of 4.3 ms versus SSM users. By the end of $t = 100$ s, DSM will have only seen their total service interruption end at 35.8 ms, whereas SSM will have amassed the entire 40 ms ($P[H] \approx 1.0$), causing a fluctuation in service delivery between the two types of users. All these results suggest that a smoother and lower-latency user experience will prevail for DSM users during this period of analysis compared to SSM.*

Metric 4: Connection Stability $CS(t) = 1 - P[H(t)]$ — **Stability Metric**

What it is: Connection Stability is defined as the percentage of users that have not experienced a handover (switchover) at time t , calculated by the formula: Connection Stability = $1 - \text{Handover Rate (Probability)}$. Higher values of Connection Stability indicate that more users are still on their original and uninterrupted connections.

Why it is appropriate: *This measure converts Handover Rate (Probability) into something that is of direct interest to network operators: how many users in their network are currently using a stable connection? Thus allowing comparison of SSM (Single Site Management) with DSM (Dynamic Subscriber Management) easily and intuitively.*

Time t	M4 SSM $CS(t)$	M4 DSM $CS(t)$	DSM Advantage
$t = 30$ s	41.8%	40.3%	Nearly equal
$t = 50$ s	17.2%	22.9%	DSM better by 5.7%
$t = 60$ s	9.7%	18.0%	DSM better by 8.3%
$t = 88$ s	1.3%	12.1%	DSM better by 10.8%
$t = 100$ s	0.4%	10.5%	DSM better by 10.1%
$t = 150$ s	0.0%	7.0%	DSM better by 7.0%
$t = 200$ s	0.0%	5.1%	DSM better by 5.1%

Key finding: At $t = 100$ s, 0.4% of users using Simple Subscription Manual (SSM) maintained an unbroken connection, while 10.5% of users using Dynamic Subscription Management (DSM) maintained an unbroken connection. At $t = 200$ s, there are 5.1% of DSM users with zero handovers, compared to 0% of SSM users. This major difference is reflected on the M4 Handover Probability graph, in which the green DSM line remains consistently below the red SSM line after approximately $t = 40$ s, therefore the difference between these two lines represents the connection stability benefit that DSM provides over SSM at each given time.

Interpretation: Why Does the Randomized DSM Perform Better?

In the SSM, all the drones are moving at one speed and have the same distance to their destination which causes all of them to drift together. Also, when another drone comes close enough to a user, they will take

over the connection to that user which will require a hand-over. Hence users cannot be protected from any variation in the connections to the drones.

In the DSM, each drone has been assigned a different speed from a continuous distribution and thus some drones will be travelling quickly while drifting away, and some will be travelling slower and remaining near their starting position. The user will always connect to the drone closest to him; therefore any drone that has randomly drawn a slow speed becomes a natural anchor as it remains closest to the user longer than any of the faster drones could possibly pass it. Therefore, this controlled randomness in the speed of the drones creates a natural stabilization to the location of the people and, thus the DSM has a much larger delay for handovers than the SSM.

The proof of that property is encapsulated in Theorem 3 of the paper, and shows that the probability of a handover in the DSM is always less than that of the SSM as time approaches infinity. The results of both the SSM and DSM under the deterministic known parameters of M3 and the random parameters of M4 demonstrate this result is both strong and general.

Summary of All Four Metrics

Metric	M3 Det. SSM	M4 Ran. SSM	M4 Ran. DSM
$P[H(t)]$ at $t = 100$ s	≈ 1.00	1.00	0.90
Comm. Cost Saving (peak)	—	—	10.8% at $t = 88$ s
Latency Saving (peak)	—	—	4.3 ms at $t = 88$ s
$CS(t)$ at $t = 100$ s	$\approx 0.7\%$	0.4%	10.5%
$t_{90\%}$ (90% HO)	69 s	60 s	104 s
Theory-sim match	Very tight	Good	Good

Overall conclusion: Across network-relevant metrics — lower handover probability, lower communication cost, lower total latency, and higher temporal connection stability — in every time (greater than or equal to 40 s). The randomized parameter configuration in M4 indicates that these benefits are applicable not only to the transient conditions examined in M3 and this paper, but also within a wide variety of potential deployments.

Scribe Question 6: Insights and Remaining Challenges

What did we learn from this experiment?

- **The theory works for randomized parameters too.** The theory had been validated only with the paper's predetermined values in M3 and then again in M4 through the MSE between the DSM simulation and the theoretical lower bound through a completely different set of parameters to give an MSE of only 6.2×10^{-6} . This proves the generality and robustness of the paper's formulas.
- **Higher drone density speeds up handovers.** M4 drew $\lambda_0 = 1.44/\text{km}^2$ vs M3's $1.0/\text{km}^2$. This alone moved $t_{50\%}$ from 38 s to 25 s for SSM. More drones eligible for one of them as the closest drone means that handovers can occur earlier and this is shown in M2 as well. The more density of drones available, the more likely it is that you will be switching drones.
- **DSM performs better on all four metrics.** Lower probability of having a handover, less communication costs, lower cumulative latencies and better connection stability some $t > 40$ s. If the user requires a stable connection for extended periods of time, then using variable-speed drones proves to be a better alternative.
- **Both models behave similarly at early times.** Both the models behaved similarly in the short time span ($t < 40$ s). The M3 and M4 graphs show that the SSM and DSM curves begin pretty much close together, but it is not until about $t \approx 40$ seconds that the curves significantly diverge. The advantage to the model selection for very short lengths of time does not have much bearing on the total time a user will be using a variable-speed model or a fixed-speed model.

When does the randomized method exceed or fall short of expectation?

- The benefit of the randomized algorithm is that it examines the theory for many different input parameter settings and establishes that the conclusions are not restricted to a single input parameter setting, which could never be shown using only the deterministic baseline.
- The disadvantage of the randomized method is that it creates a higher degree of noise through fewer iterations; Using 1,000,000 iterations in the M3 simulation provided handover coordinate values that matched almost perfectly with theoretical values. The M4 simulation, which only used 10,000 iterations, added a greater variance to the handover coordinate values; Still, the agreement was reasonable overall.
- One remaining limitation is that the simulation uses a circular area with a radius of 30 km, whereas the mathematical model assumes an infinite plane. Drones that wander outside of the predetermined limits decrease the effective density as time increases, creating a small mismatch between the theoretical and simulated values that cannot be accounted for mathematically.

- A second limitation is that the lower bound for the DSM algorithm becomes looser as it becomes larger (i.e., time). The dotted purple Rayleigh lower bound from the M4 Handover Probability graph falls below the DSM simulation curve for large t . This lower bound was based on neglecting the time correlation of drone positions. Finding a tighter lower bound as t increases remains an open issue in this paper.
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End of Milestone 4 Scribe